

Xuyang Wang

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Profile

Applied AI and systems engineer (M.S. Robotics, JHU; B.S. CS & Data Science, NYU, GPA 3.77/4.0). Builds Python/C++ systems that automate ML workflows, integrate models into pipelines, and quantify impact with reproducible metrics.

Education

Johns Hopkins University

M.S. Robotics, Mechanical Engineering

Baltimore, MD

Sep 2025 – May 2027

New York University

B.S. Computer Science & Data Science

New York, NY

Sep 2021 – May 2025

Technical Skills

Languages: Python, C++, C, C#, SQL

AI/ML: PyTorch, TensorFlow, Gymnasium, scikit-learn, LLM prompts, QLoRA fine-tuning, experiment automation

Systems: Linux, Git, ROS/cisst, Geomagic/Phantom, Maxon EPOS, teleoperation, data pipelines, debugging, visualization

Work Experience

EmotiBot

AI Engineering Intern

Shanghai, China

Jul 2023 – Aug 2023

- Owned LLM prompt and evaluation workflows for teaching, testing, interviewing, and automated scoring.
- Built Python data pipelines to collect, clean, and structure corpora for supervised fine-tuning.
- Implemented a QLoRA fine-tuning workflow for customer-service automation and content generation.
- Validated end-to-end model behavior through testing and debugging on an iOS learning product.

Project Experience

CIS II: EyeRobot and Eye Snake Teleoperation

Jan 2026 – Present

Software Integration Lead

- Architected ROS/cisst control for unified EyeRobot (SHER 2.1) and Eye Snake (I2RIS) teleoperation under one operator loop.
- Implemented C++ teleoperation in `robotTask.cpp` with Geomagic/Phantom ROS and Maxon EPOS packages.
- Delivered `SHER_I2RIS_teleoperate` launch workflows; fixed mapping, stability, joint limits, and build integration issues.

Move–Aim–Shoot RL Benchmark

Jan 2025 – Present

Lead Developer

- Designed a reproducible 2D RL benchmark with 12 maps, self-play, and partial observability (laser scans).
- Built Gymnasium environments and shared training/evaluation APIs for hybrid and continuous actions.
- Implemented feedforward PPO, recurrent PPO with privileged critic (CTDE), asymmetric distillation, and SAC.
- Automated fixed-teacher evaluation with multi-seed mean $\pm$ 95% CI and robustness tests ($\sim$ 80–93% win rate in $\sim$ 5k steps).

Multitask RL with OOD Monitoring

Sep 2023 – Dec 2023

ML Engineer

- Detected OOD policy drift during fine-tuning; defined distance metrics/thresholds and built training visualization tooling.